DEPARTMENT OF PHYSICS
INDIAN INSTITUTE OF TECHNOLOGY, MADRAS

PH1020 Physics II

Problem Set 3

Conductors, Laplace's equation, and Multipole expansion Date: 16 Jan 2025

1. Two spherical cavities, of radii a and b, are hollowed out from the interior of a (neutral) conducting sphere of radius R as shown in the figure 1 below. At the center of each cavity respective point charge q_a and q_b are placed.
   (a) Find the surface charge densities σ_a, σ_b and σ_R.
   (b) What is the field outside the conductor?
   (c) What is the field within each cavity?
   (d) What is the force on q_a and q_b ?
   (e) Which of these answers would change if a third charge q_c, were brought near the conductor?

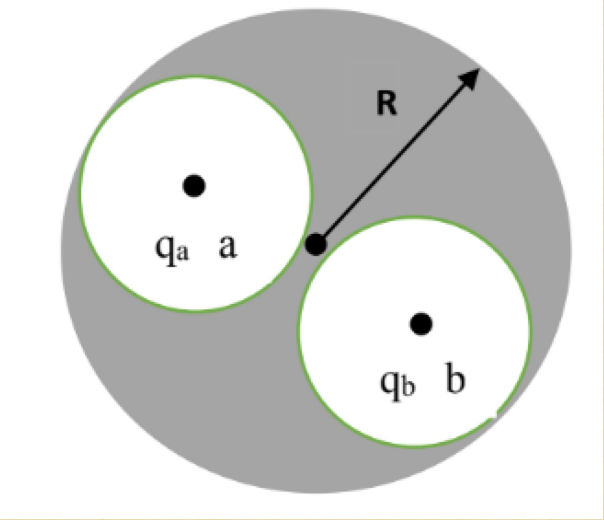


Figure 1: Neutral conducting sphere with two spherical cavities.

2. Two large metal plates, each of area A, are separated by a distance d. If we put charge Q on each plate, what will be the electrostatic pressure on the plates?
3. Find the capacitance per unit length of two coaxial metallic cylindrical tubes of radii a and b, where $b > a$.
4. The computation of the electrostatic potential $\phi(\mathbf{r})$ reduces to solving Laplace's equation, $\nabla^2\phi = 0$, subject to boundary conditions. Consider the following situations:
   (a) Consider a situation with cylindrical symmetry and translational symmetry along the z-direction. In such cases, the potential is independent of the coordinates φ and z, i.e., $\phi = \phi(\varrho)$. Show that the most general solution to Laplace's equation in this case takes the form $\phi = A \log \varrho + B$ for suitable

constants A and B. Determine A and B using the following boundary conditions:

$$\phi(\varrho)\big|_{\varrho=a} = V_0, \quad \phi(\varrho)\big|_{\varrho=b} = 0, \quad \text{for } a > b > 0.$$

Take the region of interest to be $b \leq \varrho \leq a$. Compute the electric field for this electrostatic potential. Plot lines of equipotential and electric lines of force.

(b) Consider two (infinite) metal plates parallel to each other and separated by a distance d. One plate is grounded, and the other is kept at V_0. Choose the direction of separation to be the y-axis. Solve Laplace's equation in the region between the two plates subject to the given boundary conditions. Compute the electric field for this electrostatic potential. Plot lines of equipotential and electric lines of force.

(c) Consider a grounded (i.e., at zero potential) infinitely long metal sheet lying in the $z = 0$ plane. A point charge $+Q$ is located at $z = a > 0$. Verify that:

$$\phi(x, y, z) = \frac{1}{4\pi\epsilon_0}\left(\frac{Q}{\sqrt{x^2 + y^2 + (z-a)^2}} - \frac{Q}{\sqrt{x^2 + y^2 + (z+a)^2}}\right),$$

solves Laplace's equation at all points in the region $z \geq 0$ except for the location of the charge, as well as satisfies the boundary conditions. If possible, interpret the second term in the above expression in terms of an image charge located somewhere.

5. A *point* dipole moment $\mathbf{p}$ is defined as follows: the separation $2a$ between the positive charge q and the negative charge $-q$ tends to zero, while q tends to infinity, such that $p \equiv 2aq$ is a finite positive number.

(a) Show that the electric field at any point $\mathbf{r}$ due to a point dipole at the origin is given by:

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0}\frac{3(\mathbf{p}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{p}}{r^3}.$$

(b) Suppose the direction of the dipole moment $\mathbf{p}$ makes an angle θ_p with the z-axis. Calculate the electric flux through the upper hemisphere of a spherical surface of radius R centered at the origin.

6. (a) Show that if the total charge of a system is zero, the dipole moment does not depend on the location of the origin of the coordinates.

(b) If a charge distribution has a non-zero monopole moment, show that it is possible to find an origin about which the dipole moment is zero.

(c) Show that the quadrupole moment $\mathbf{Q}$ of a charge distribution does not depend on the location of the origin, if the monopole and dipole moments of the charge distribution are both zero.

7. Show that for a spherically symmetric charge distribution, the dipole, quadrupole, and all higher moments about the center of the distribution are identically zero.

8. Three point charges $+q$, $+2q$, and $-4q$ are held fixed at points with Cartesian coordinates $(0, 0, d)$, $(-d, 0, d)$, and (d, d, d) respectively, in a Cartesian coordinate system. Calculate the quadrupole moment of the charge distribution about the point where the dipole moment vanishes.

9. Find the monopole, dipole, and quadrupole moments about the origin for the following charge distributions:

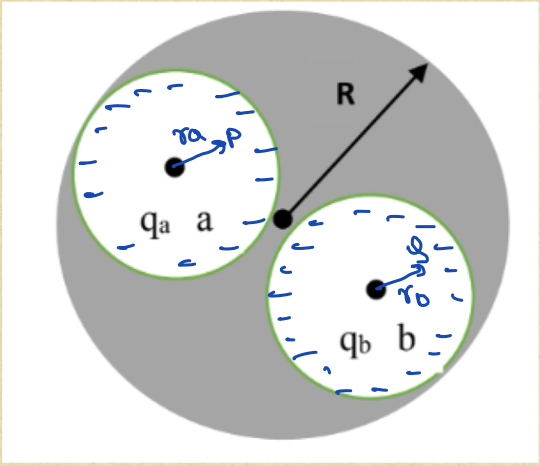
   (a) A line charge of constant charge density λ lying in the first quadrant of the xy-plane with one end at the origin and making an angle φ_0 with the positive x-axis.

   (b) A solid cylinder of height H and radius R with its geometrical center at the origin, having a uniform volume charge density ρ_0, and its axis along the z-axis.

1. Two spherical cavities, of radii a and b, are hollowed out from the interior of a (neutral) conducting sphere of radius R as shown in the figure 1 below. At the center of each cavity respective point charge q_a and q_b are placed.
   (a) Find the surface charge densities σ_a, σ_b and σ_R. $-q_a/4\pi a^2$ $-q_b/4\pi b^2$, $(q_a+q_b)/4\pi R^2$
   (b) What is the field outside the conductor? $\hookrightarrow \frac{1}{4\pi\varepsilon_0}\left(\frac{q_a+q_b}{r^2}\right)\hat{r}$
   (c) What is the field within each cavity?
   (d) What is the force on q_a and q_b ? $\hookrightarrow$ zero
   (e) Which of these answers would change if a third charge q_c, were brought near the conductor? σ_R, $E_{outside}$

$\frac{1}{4\pi\varepsilon_0}\frac{q_A}{r_a^2}\hat{r}_a$

$\frac{1}{4\pi\varepsilon_0}\frac{q_b}{r_b^2}\hat{r}_b$

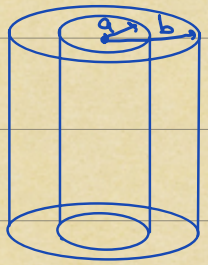


2. Two large metal plates, each of area A, are separated by a distance d. If we put charge Q on each plate, what will be the electrostatic pressure on the plates?

$E = \sigma/2\varepsilon_0$

$P = F/A = qE/A = \sigma^2/2\varepsilon_0$

3. Find the capacitance per unit length of two coaxial metallic cylindrical tubes of radii a and b, where $b > a$.



$E \times 2\pi r l = q/\varepsilon_0$

$\Rightarrow E = \frac{q}{2\pi\varepsilon_0 r l}$

$V = -\int_a^b E\,dr = \frac{q}{2\pi\varepsilon_0 l}\ln(b/a) = q/C$

$\Rightarrow C = \frac{2\pi\varepsilon_0 l}{\ln(b/a)}$

(a) Consider a situation with cylindrical symmetry and translational symmetry along the z-direction. In such cases, the potential is independent of the coordinates φ and z, i.e., $\phi = \phi(\varrho)$. Show that the most general solution to Laplace's equation in this case takes the form $\phi = A \log \varrho + B$ for suitable

$$\nabla^2 V = 0$$

$$\vec{\nabla} V = \frac{dV}{dr}\hat{r} + \frac{1}{r}\frac{dV}{d\theta}\hat{\theta} + \frac{dV}{dz}\hat{z}$$

$$\vec{\nabla}^2 V = \frac{1}{r}\frac{d}{dr}\left(r\frac{dV}{dr}\right) + 1/r^2 \frac{d^2V}{d\theta^2} + \frac{d^2V}{dz^2} = 0$$

$$\Rightarrow \frac{d}{dr}\left(r\frac{dV}{dr}\right) = 0$$

$$\Rightarrow r\frac{dV}{dr} = A$$

$$dV = A/r\, dr$$

$$V = A\ln(r) + B$$

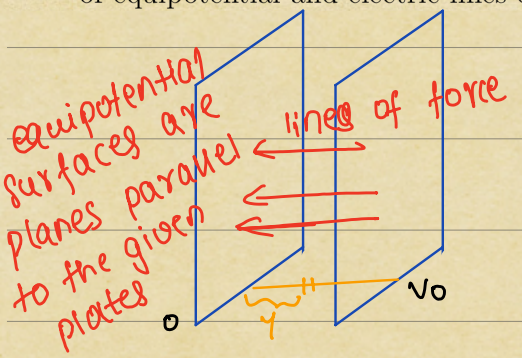
$$V_0 = A\ln a + B \qquad \frac{V_0}{\ln a - \ln b} = A \;,\; B = \frac{-V_0 \ln b}{\ln a - \ln b}$$

$$0 = A\ln b + B$$

$$\Rightarrow V = \frac{V_0[\ln r - \ln b]}{\ln a - \ln b} = \frac{V_0 \ln(r/b)}{\ln(a/b)}$$

$$E = \frac{-V_0\,\hat{r}}{r\ln(a/b)}$$

(b) Consider two (infinite) metal plates parallel to each other and separated by a distance d. One plate is grounded, and the other is kept at V_0. Choose the direction of separation to be the y-axis. Solve Laplace's equation in the region between the two plates subject to the given boundary conditions. Compute the electric field for this electrostatic potential. Plot lines of equipotential and electric lines of force.



$$\frac{d^2V}{dy^2} = 0$$

$$\Rightarrow \frac{dV}{dy} = a$$

$$\Rightarrow V = ay + b$$

$$\Rightarrow V = \frac{V_0 y}{d}$$

$$E_y = -\frac{V_0}{d}$$

(c) Consider a grounded (i.e., at zero potential) infinitely long metal sheet lying in the $z = 0$ plane. A point charge $+Q$ is located at $z = a > 0$. Verify that:

$$\phi(x, y, z) = \frac{1}{4\pi\epsilon_0}\left(\frac{Q}{\sqrt{x^2 + y^2 + (z-a)^2}} - \frac{Q}{\sqrt{x^2 + y^2 + (z+a)^2}}\right),$$

solves Laplace's equation at all points in the region $z \geq 0$ except for the location of the charge, as well as satisfies the boundary conditions. If possible, interpret the second term in the above expression in terms of an image charge located somewhere.

$$\nabla^2\phi = 0$$

$$\nabla^2(1/r) = 4\pi\delta(r)$$

$$\nabla^2\phi = -\frac{1}{4\pi\epsilon_0}\, 4\pi Q\left[\delta(x)\delta(y)\delta(z-a) - \delta(x)\delta(y)\delta(z+a)\right]$$

$$= \frac{Q}{\epsilon_0}\delta(x)\delta(y)\left[\delta(z-a) - \delta(z+a)\right]$$

0 everywhere except at $z = a$

5. A *point* dipole moment $\mathbf{p}$ is defined as follows: the separation $2a$ between the positive charge q and the negative charge $-q$ tends to zero, while q tends to infinity, such that $p \equiv 2aq$ is a finite positive number.

   (a) Show that the electric field at any point $\mathbf{r}$ due to a point dipole at the origin is given by:

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0}\frac{3(\mathbf{p}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{p}}{r^3}.$$

   (b) Suppose the direction of the dipole moment $\mathbf{p}$ makes an angle θ_p with the z-axis. Calculate the electric flux through the upper hemisphere of a spherical surface of radius R centered at the origin.

(a) $\phi(r) = \frac{p\cos\theta}{4\pi\varepsilon_0 r^2}$

$\vec{E} = -\nabla(\phi) = -\frac{d\phi}{dr} - \frac{1}{r}\frac{d\phi}{d\theta}$

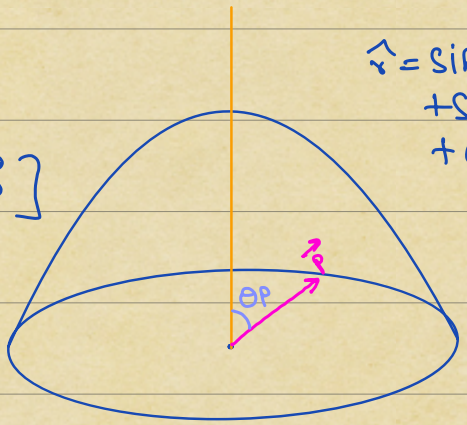
$= \frac{p}{4\pi\varepsilon_0 r^3}[2\cos\theta\,\hat{r} + \sin\theta\,\hat{\theta}]$

$(\vec{p}\cdot\hat{r})\hat{r} = p\cos\theta\,\hat{r}$

$\vec{p} = p\cos\theta + p\sin\theta$

(b) $E = \frac{1}{4\pi\varepsilon_0 r^3}[3(\vec{p}\cdot\hat{r})\hat{r} - \vec{p}]$

$\hat{r} = \sin\theta\cos\phi\,\hat{x} + \sin\theta\sin\phi\,\hat{y} + \cos\theta\,\hat{z}$



$\phi = \int \vec{E}\cdot d\vec{s}$

$\vec{p}\cdot\hat{r} = p_x\sin\theta\cos\phi + p_y\sin\theta\sin\phi + p_z\cos\theta$ $\quad ds = R^2\sin\theta\,d\theta\,d\phi\,\hat{r}$

$\phi = \frac{p}{2\pi\varepsilon_0 R}\int \sin^2\theta\cos\phi\,d\theta\,d\phi + \int \sin^2\theta\sin\phi\,d\theta\,d\phi + \int \cos\theta\sin\theta\,d\theta d\phi$

$= \frac{p_z}{4\pi\varepsilon_0 R} \times 2 \times \pi = \frac{p\cos\theta_p}{2\varepsilon_0 R}$

6. (a) Show that if the total charge of a system is zero, the dipole moment does not depend on the location of the origin of the coordinates.

(b) If a charge distribution has a non-zero monopole moment, show that it is possible to find an origin about which the dipole moment is zero.

(c) Show that the quadrupole moment **Q** of a charge distribution does not depend on the location of the origin, if the monopole and dipole moments of the charge distribution are both zero.

$r_i \rightarrow$ initial position

a) $\vec{r}_{new} = \vec{r}_i - \vec{a}$

$\vec{p} = \sum q_i \vec{r}_{new} = \sum q_i \vec{r}_i - \sum q_i \vec{a} = \vec{p} - 0 = \vec{p}$

b) $\sum q_i \neq 0$

$\vec{a} = \frac{\sum q_i \vec{r}_i}{\sum q_i}$

$x_n = x - a_x$
$y_n = y - a_y$
$z_n = z - a_z$

c) $Q_{ij} = \int (3 r_i r_j - r^2 \delta_{ij}) \rho \, dV$

$(Q_{xx})_{new} = \int \rho \, dV \, [2x_{new}^2 - y_{new}^2 - z_{new}^2]$

$= \int \rho \, dV \, [2x^2 - y^2 - z^2] + \int \rho \, dV \, (2a_x^2 - a_y^2 - a_z^2)$

$+ \int \rho \, dV \, [-4x a_x + 2y a_y + 2z a_z]$

$= (Q_{xx})_{old} + (2a_x^2 - a_y^2 - a_z^2) Q_{total} + (-4 a_x p_x + 2 a_y p_y + 2 a_z p_z)$

$\Rightarrow (Q_{xx})_{new} = (Q_{xx})_{old}$

$(Q_{xy})_{new} = \int (3 x_{new} y_{new}) \rho \, dV = \int 3 (x - a_x)(y - a_y) \rho \, dV$

$= \int 3xy \rho \, dV + 3 a_x a_y \int \rho \, dV - 3 p_x a_y - 3 p_y a_x$

$= (Q_{xy})_{old}$

$\Rightarrow Q$ doesn't change

7. Show that for a spherically symmetric charge distribution, the dipole, quadrupole, and all higher moments about the center of the distribution are identically zero.

for spherically symmetric distribution,

$\phi(r) = \frac{1}{4\pi\varepsilon_0 r} \int \rho \, dv \rightarrow$ monopole term

$\Rightarrow$ other terms are zero

8. Three point charges $+q$, $+2q$, and $-4q$ are held fixed at points with Cartesian coordinates $(0, 0, d)$, $(-d, 0, d)$, and (d, d, d) respectively, in a Cartesian coordinate system. Calculate the quadrupole moment of the charge distribution about the point where the dipole moment vanishes.

$\vec{a} = \frac{-6qd, -4qd, -qd}{-q} = (6d, 4d, d)$

q_i	x_i	y_i	z_i	x_i'	y_i'	z_i'
q	0	0	d	$-6d$	$-4d$	0
$2q$	$-d$	0	d	$-7d$	$-4d$	0
$-4q$	d	d	d	$-5d$	$-3d$	0

$Q_{ij} = \sum [3x_i x_j - r^2 \delta_{ij}] q_i$

$Q_{xx} = \sum [3x_x^2 - x_y^2 - x_z^2] q_i = q[56]d^2 + 2q[52d^2] - 4q[41d^2]$

$Q_{yy} = q[-4d^2] + 2q[-7d^2] - 4q[-7d^2] = -10qd^2$

$Q_{zz} = +q[-52d^2] + 2q[-65d^2] - 4q[34d^2] = -460qd^2$

$Q_{xy} = Q_{yx} = 72qd^2 + 168qd^2 - 180qd^2 = 60qd^2$

$Q_{yz} = Q_{zy} = Q_{xz} = Q_{zx} = 0$

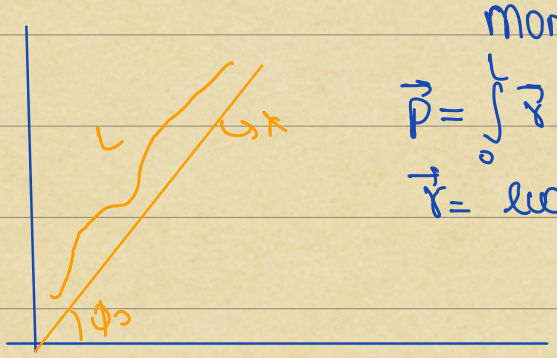
$Q_{xx} + Q_{yy} + Q_{zz} = 0$

9. Find the monopole, dipole, and quadrupole moments about the origin for the following charge distributions:

   (a) A line charge of constant charge density λ lying in the first quadrant of the xy-plane with one end at the origin and making an angle φ_0 with the positive x-axis.

   (b) A solid cylinder of height H and radius R with its geometrical center at the origin, having a uniform volume charge density ρ_0, and its axis along the z-axis.

(a)



Monopole $= \lambda L$

$$\vec{p} = \int_0^L \vec{r}\,(\lambda\, d\vec{\ell})$$

$$\vec{r} = \ell\cos\phi\,\hat{x} + \ell\sin\phi\,\hat{y}$$

$$\vec{p} = \frac{\lambda L^2}{2}(\cos\phi\,\hat{x} + \sin\phi\,\hat{y})$$

$$Q_{xx} = \int_0^L (3x^2 - r^2)\lambda\, d\ell = \int_0^L (3\ell^2\cos^2\phi - \ell^2)\lambda\, d\ell$$

$$= \frac{\lambda L^3}{3}(3\cos^2\phi - 1)$$

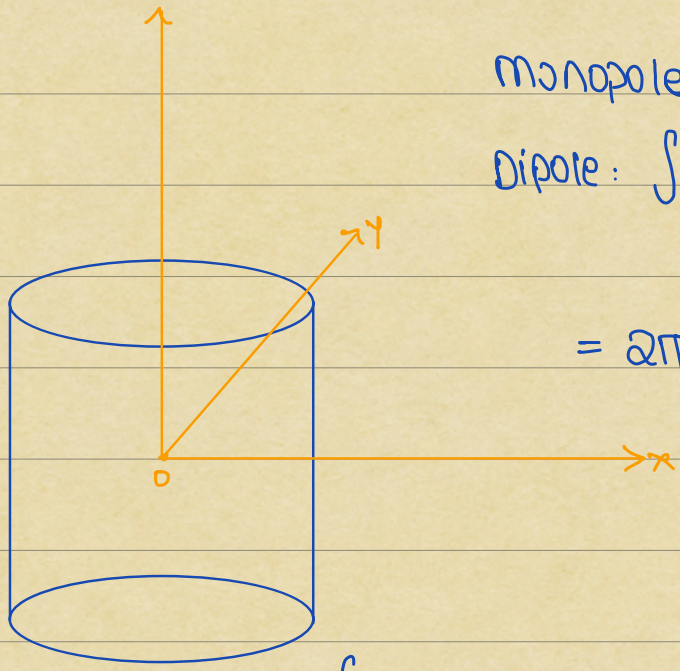
$$Q_{yy} = \frac{\lambda L^3}{3}(3\sin^2\phi - 1)$$

$$Q_{zz} = -\frac{\lambda L^3}{3}$$

$$Q_{xy} = \int_0^L 3\ell^2 \sin\phi\cos\phi\,\lambda\, d\ell = \lambda L^3 \sin\phi\cos\phi = Q_{yx}$$

$$Q_{\text{others}} = 0$$

(b)



Monopole : $\rho_0 \times \pi R^2 H$

Dipole : $\int \rho_0 \cdot \frac{r \cdot dr \cdot d\theta \, dz}{\sqrt{r^2+z^2}}$

$= 2\pi\rho_0 \int_{-H/2}^{H/2} \int_0^r \frac{r \, dr \, dz}{\sqrt{r^2+z^2}}$

$= 0$

Quadrupole: $Q_{zz} = \int (2z^2 - r^2) \rho \, r \, dr \, d\theta \, dz$

$= 2\pi\rho \int \left(r^2 z^2 - \frac{r^4}{4} \right) dz$

$= 2\pi\rho \left[R^2 \frac{z^3}{3} - \frac{R^4 z}{4} \right]$ $\quad z \to -H/2 \text{ to } H/2$

$= 2\pi\rho \left[\frac{R^2}{3} \frac{H^3}{4} - \frac{R^4}{4} H \right]$

$= \frac{\pi\rho R^2 H}{2} \left[\frac{H^2 - 3R^2}{3} \right] = \frac{Q}{6} [H^2 - 3R^2]$

$Q_{xx} = Q_{yy} = -\frac{Q}{12} [H^2 - 3R^2]$

$Q_{ij} \; (i \neq j) = 0$

DEPARTMENT OF PHYSICS
INDIAN INSTITUTE OF TECHNOLOGY, MADRAS
PH1020 Physics II
Problem Set 3 - Solutions

Conductors, Laplace's equation, and Multipole expansion Date: 10 Feb 2025

1. Problem-1

Question: Two spherical cavities, of radii a and b, are hollowed out from the interior of a (neutral) conducting sphere of radius R as shown in the figure 1 below. At the center of each cavity respective point charge q_a and q_b are placed.
(a) Find the surface charge densities σ_a, σ_b and σ_R.
(b) What is the field outside the conductor?
(c) What is the field within each cavity?
(d) What is the force on q_a and q_b ?
(e) Which of these answers would change if a third charge q_c, were brought near the conductor?

Solution:

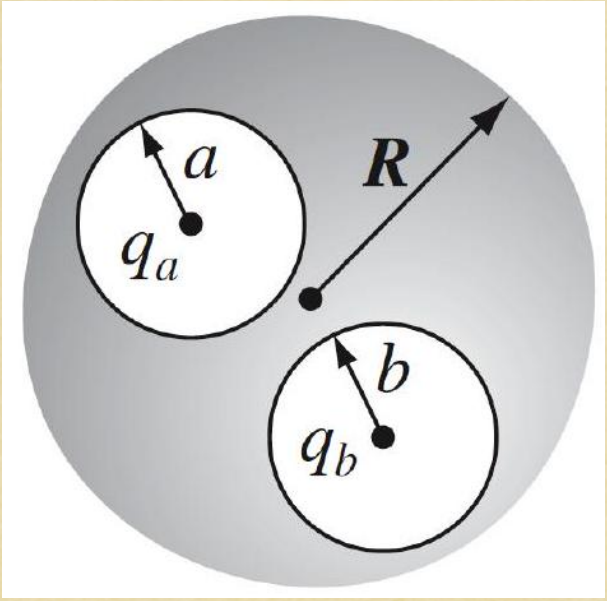


Figure 1: Neutral conducting sphere with two spherical cavities.

(a) Following the arguments in section (2.5.1) and (2.5.2) of Griffiths book

$$\sigma_a = -\frac{q_a}{4\pi a^2} \quad \sigma_b = -\frac{q_b}{4\pi b^2} \quad \sigma_R = \frac{q_a + q_b}{4\pi R^2}$$

(b) Electric field at a point $\mathbf{r}(> R)$ from the center of the sphere

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0}\frac{q_a + q_b}{|\mathbf{r}|^3}\mathbf{r}$$

(c) Field inside the cavity with radius a, and at $\mathbf{r}_a$ from the center of the cavity,

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0}\frac{q_a}{|\mathbf{r}_a|^3}\mathbf{r}_a$$

Field inside the cavity with radius b, and at $\mathbf{r}_b$ from the center of the cavity,

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0}\frac{q_b}{|\mathbf{r}_b|^3}\mathbf{r}_b$$

(d) As both the charges q_a and q_b are in the cavity inside a conductor, the force on each of the charges will be *zero*.

(e) If another charge q_c was brought near the conductor then, following the arguments in section (2.5.1) and (2.5.2) of Griffiths book, the surface charge density of the outer surface σ_R will change (and hence the electric field out side the conductor will also change) to make sure the field inside the conductor is zero. Whereas, the surface charge density of the cavity i.e. σ_a and σ_b will not change, neither the force on each charge.

2. Problem-2

Question: Two large metal plates, each of area A, are separated by a distance d. If we put charge Q on each plate, what will be the electrostatic pressure on the plates?
Solution:

Let the electric fields produced by plate 1 and 2 are $\boldsymbol{E}_1$ and $\boldsymbol{E}_2$ respectively (See Fig. 2). The magnitude can be written as, $|\boldsymbol{E}_1| = |\boldsymbol{E}_2| = \frac{\sigma}{2\epsilon_0}$ (where $\sigma = \frac{Q}{A}$). The electric field between the large conducting plates (region II) is zero and $\frac{\sigma}{\epsilon_0}$ outside (Region I and III). Hence, the electrostatic pressure

$$P = \frac{\sigma^2}{2\epsilon_0}$$

3. Problem-3

Question: Find the capacitance per unit length of two coaxial metallic cylindrical tubes of radii a and b, where $b > a$.
Solution:

Assume the charge per unit length is given by λ. Exploiting the cylindrical symmetry and using Gauss's law we can write

$$\int E \cdot dS = E2\pi\rho L = \frac{Q_{\text{enc}}}{\epsilon_0} = \lambda L$$

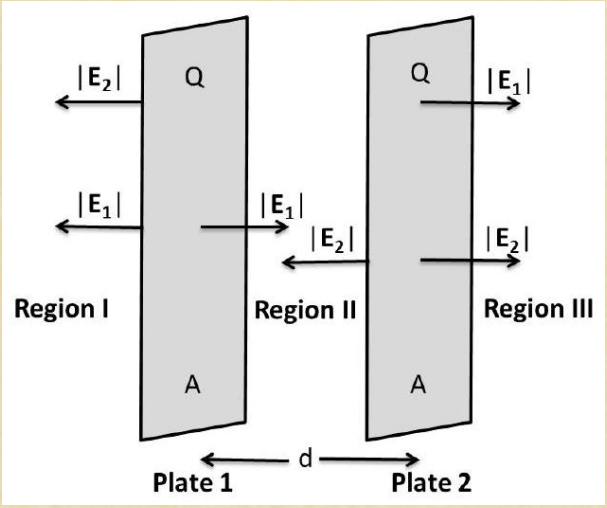


Figure 2: .

we get $E_\rho = \frac{\lambda}{2\pi\epsilon_0}\frac{1}{\rho}\hat{e}_\rho$. To evaluate the potential difference between the cylinders we use

$$V(b) - V(a) = -\int_a^b E \cdot \ell = -\frac{\lambda}{2\pi\epsilon_0}\int_b^a \frac{1}{\rho} d\rho = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{b}{a}\right).$$

Thus, the capacitance is

$$C = Q/V = \frac{2\pi\epsilon_0 L}{\ln\left(\frac{b}{a}\right)}$$

4. Problem-4

Question: The computation of the electrostatic potential $\phi(\mathbf{r})$ reduces to solving Laplace's equation, $\nabla^2\phi = 0$, subject to boundary conditions. Consider the following situations:

(a) Consider a situation with cylindrical symmetry and translational symmetry along the z-direction. In such cases, the potential is independent of the coordinates φ and z, i.e., $\phi = \phi(\varrho)$. Show that the most general solution to Laplace's equation in this case takes the form $\phi = A \log \varrho + B$ for suitable constants A and B. Determine A and B using the following boundary conditions:

$$\phi(\varrho)\big|_{\varrho=a} = V_0, \quad \phi(\varrho)\big|_{\varrho=b} = 0, \quad \text{for } a > b > 0.$$

Take the region of interest to be $b \le \varrho \le a$. Compute the electric field for this electrostatic potential. Plot lines of equipotential and electric lines of force.

(b) Consider two (infinite) metal plates parallel to each other and separated by a distance d. One plate is grounded, and the other is kept at V_0. Choose the direction of separation to be the y-axis. Solve Laplace's equation in the region between the two plates subject to the given boundary conditions. Compute the electric field for this electrostatic potential. Plot lines of equipotential and electric lines of force.

(c) Consider a grounded (i.e., at zero potential) infinitely long metal sheet lying in the $z = 0$ plane. A point charge $+Q$ is located at $z = a > 0$. Verify that:

$$\phi(x, y, z) = \frac{1}{4\pi\epsilon_0}\left(\frac{Q}{\sqrt{x^2 + y^2 + (z-a)^2}} - \frac{Q}{\sqrt{x^2 + y^2 + (z+a)^2}}\right),$$

solves Laplace's equation at all points in the region $z \geq 0$ except for the location of the charge, as well as satisfies the boundary conditions. If possible, interpret the second term in the above expression in terms of an image charge located somewhere.

Solution:

(a)
In cylindrical coordinate system general solution is given by,

$$\phi(\varrho) = A \log \varrho + B$$

Here boundary conditions are given as

$$\phi(\varrho)\big|_{\varrho=a} = V_0, \quad \phi(\varrho)\big|_{\varrho=b} = 0, \quad \text{for } a > b > 0.$$

Therefore, for $b \leq \varrho \leq a$

$$\phi(\varrho) = V_0 \frac{\log(\varrho/b)}{\log(a/b)}$$
$$\mathbf{E} = -\frac{V_0}{\varrho \log(a/b)} \hat{\varrho}.$$

Plot of lines of equipotential and electric lines of force are shown in Fig. 3.

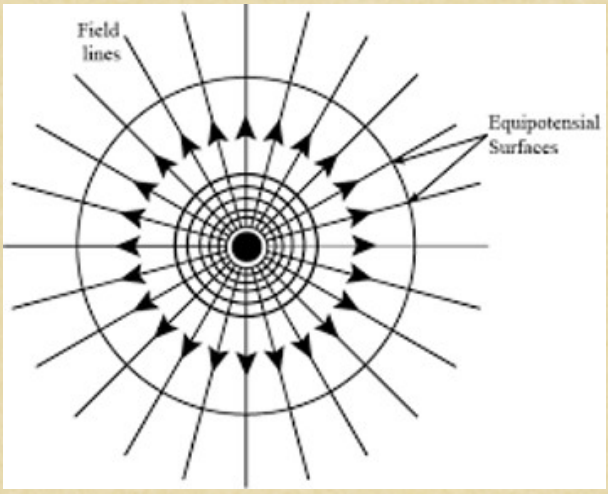


Figure 3: Lines of equipotential and electric lines of force.

(b) This is a 1-D case where the Laplace's equation is

$$\frac{d^2V}{dy^2} = 0$$

The general solution is given by

$$V(y) = my + b$$

Here the constants m and b can be determined from the boundary conditions.

$$V(y = 0) = 0, V(y = d) = V_0$$

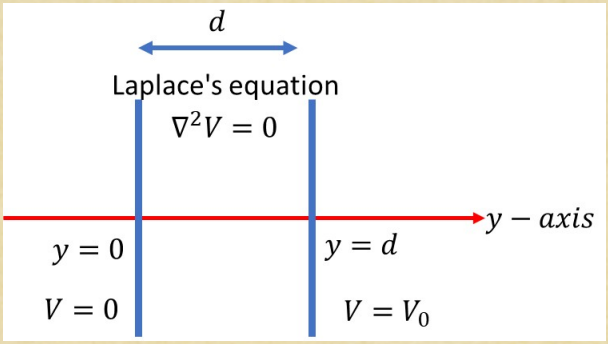


Figure 4: Two (infinite) metal plates (along the y-axis) parallel to each other and separated by a distance d. One plate is grounded, and the other is kept at V_0.

The final solution (for $0 \leq y \leq d$) is

$$V(y) = V_0 \frac{y}{d}$$
$$\mathbf{E} = -\nabla V$$
$$\mathbf{E} = -\frac{V_0}{d}\hat{\mathbf{y}}$$

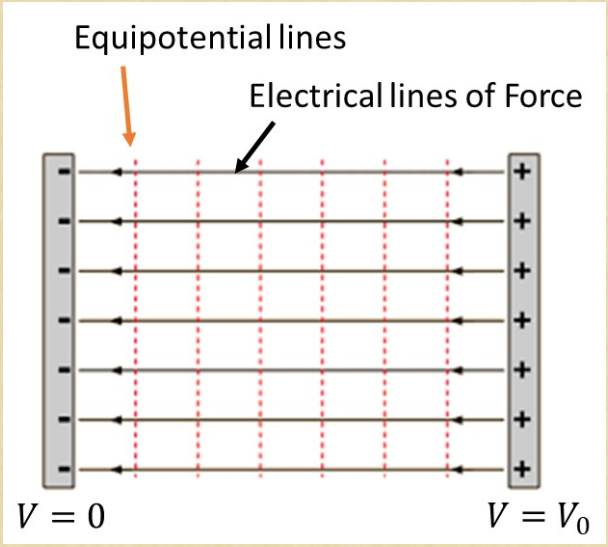


Figure 5: Lines of equipotential and electric lines of force

(c) It is straightforward to check that the given potential satisfies Laplace's equation.

$$\phi(x, y, z) = \frac{1}{4\pi\epsilon_0}\left(\frac{Q}{\sqrt{x^2 + y^2 + (z-a)^2}} - \frac{Q}{\sqrt{x^2 + y^2 + (z+a)^2}}\right)$$

The boundary conditions are that the potential vanishes for $z = 0$ and goes to zero as $z \to \infty$. Again these can be checked easily and hence no details are necessary.

Note: It is important to note that the above potential is identical to the the potential in all space i.e., in $\mathbb{R}^3$ with two charges, $+Q$ located at $(0, 0, a)$ and another charge

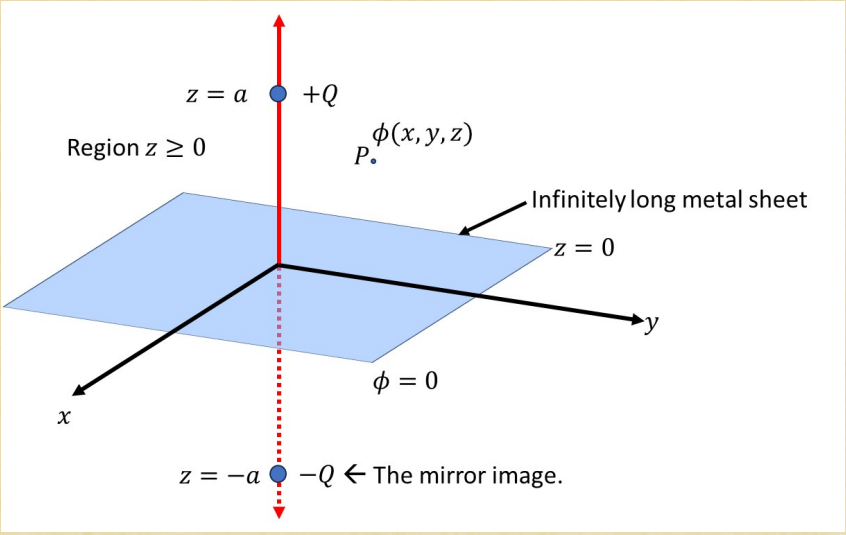


Figure 6: Grounded infinitely long metal sheet lying in the $z = 0$ plane with a point charge at $z = a$ and image charge at $z = -a$.

$-Q$ located at $(0, 0, -a)$. The location of the second charge is the **mirror image** of the location of the first charge if there were a mirror at $z = 0$. It is important to notice that the image is located outside the region of interest in our problem i.e., $z > 0$. This method of solving Laplace's equation is called the **method of images**.

A fun exercise is to ask a similar question (for the potential due to a point charge located say at $(0, a, b)$) but now with an additional grounded plate at $y = 0$ and the region of interest is now $y > 0$ and $z > 0$. How many image charges will you need?

5. Problem-5

Question: A *point* dipole moment $\mathbf{p}$ is defined as follows: the separation $2a$ between the positive charge q and the negative charge $-q$ tends to zero, while q tends to infinity, such that $p \equiv 2aq$ is a finite positive number.

(a) Show that the electric field at any point $\mathbf{r}$ due to a point dipole at the origin is given by:

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{3(\mathbf{p} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{p}}{r^3}.$$

(b) Suppose the direction of the dipole moment $\mathbf{p}$ makes an angle θ_p with the z-axis. Calculate the electric flux through the upper hemisphere of a spherical surface of radius R centered at the origin.

Solution:

(a) With no loss of generality, choose the align the dipole along the z-axis (see fig. 7). The electrostatic potential, $\phi(\mathbf{r})$, at a point L which is at a distance r from the point dipole is

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0 r^2} \boldsymbol{p} \cdot \hat{\mathbf{r}} = \frac{1}{4\pi\epsilon_0 r^2} (p \cos\theta).$$

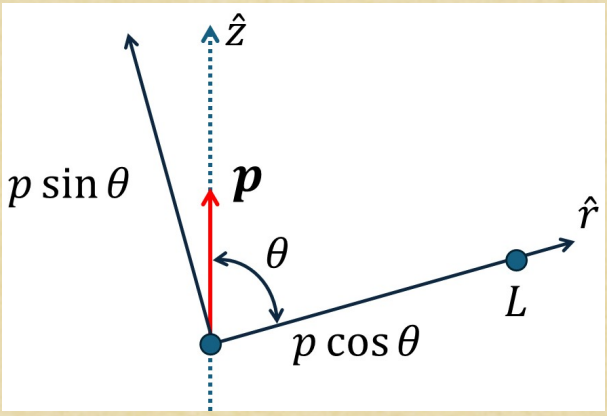


Figure 7: A point dipole aligned with the z-axis

The electric field is determined from the above potential as follows:

$$\begin{aligned}
\boldsymbol{E} &= -\boldsymbol{\nabla}\phi \\
&= \frac{p}{4\pi\epsilon_0 r^3}\left[2\cos\theta\hat{\mathbf{r}} + \sin\theta\hat{\theta}\right] \\
&= \frac{1}{4\pi\epsilon_0 r^3}\left[3\left(\boldsymbol{p}\cdot\hat{\mathbf{r}}\right)\hat{\mathbf{r}} - \left(p\cos\theta\hat{\mathbf{r}} - p\sin\theta\hat{\theta}\right)\right] \\
&= \frac{1}{4\pi\epsilon_0 r^3}\left[3\left(\boldsymbol{p}\cdot\hat{\mathbf{r}}\right)\hat{\mathbf{r}} - \boldsymbol{p}\right]
\end{aligned}$$

The last expression is independent of the coordinate system used.

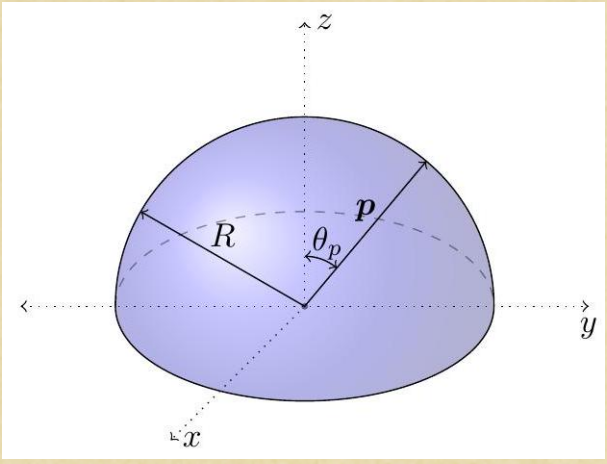


Figure 8: Calculating flux through the curved surface of upper hemisphere of a sphere.

(b) From (3a) above,

$$\begin{aligned}
\boldsymbol{E} &= \frac{1}{4\pi\epsilon_0 r^3}\left[3\left(\boldsymbol{p}\cdot\hat{\mathbf{r}}\right)\hat{\mathbf{r}} - \boldsymbol{p}\right], \\
d\boldsymbol{S} &= R^2\sin\theta d\theta d\varphi\hat{\mathbf{r}}. \\
\therefore \Phi = \int \boldsymbol{E}\cdot d\boldsymbol{S} &= \int \frac{1}{4\pi\epsilon_0 R}\left(2\boldsymbol{p}\cdot\hat{\mathbf{r}}\right)d\Omega.
\end{aligned}$$

Note that the limits on θ integration are from 0 to $\pi/2$ while the limits on φ integration are from 0 to 2π.

$$\boldsymbol{p}\cdot\hat{\mathbf{r}} = p_x\sin\theta\cos\varphi + p_y\sin\theta\sin\varphi + p_z\cos\theta$$

Since

$$\int_0^{2\pi} \cos\varphi = 0 \quad \text{and} \quad \int_0^{2\pi} \sin\varphi = 0$$

only the p_z term survives in the integration. On performing the integration, the flux is obtained to be

$$\Phi = \frac{p_z}{2\epsilon_0 R} = \frac{p\cos\theta_p}{2\epsilon_0 R}$$

6. Problem-6

Question:

(a) Show that if the total charge of a system is zero, the dipole moment does not depend on the location of the origin of the coordinates.

(b) If a charge distribution has a non-zero monopole moment, show that it is possible to find an origin about which the dipole moment is zero.

(c) Show that the quadrupole moment **Q** of a charge distribution does not depend on the location of the origin, if the monopole and dipole moments of the charge distribution are both zero.

Solution:
(a)

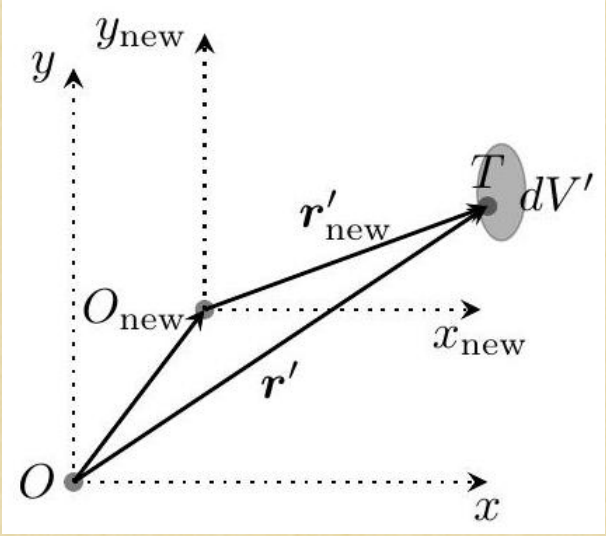


Figure 9: Shift of origin of coordinates and its effect on the dipole moment

$$\begin{aligned}
\boldsymbol{p}_{\text{new}} &= \int \varrho(T)\boldsymbol{r}'_{\text{new}}\, dV' \\
&= \int \varrho \boldsymbol{r}' dV' - a\int \varrho dV' \\
&= \boldsymbol{p} - \boldsymbol{a}Q
\end{aligned}$$

If total charge $Q = 0$, then $\boldsymbol{p}_{\text{new}} = \boldsymbol{p}$.

(b) $\boldsymbol{p}_{\text{new}}$ in (a) above will be zero if

$$\boldsymbol{p} = \boldsymbol{a}Q$$

$$\text{i.e., } \int \varrho \boldsymbol{r}' dv' = \boldsymbol{a} \int \varrho dv'$$

$$\therefore a = |\boldsymbol{a}| = \frac{\int \varrho r' dv'}{\int \varrho dv'} \tag{1}$$

Hence, the point about which the new dipole moment is zero is at a distance a from O (' a ' as given in equation 1. So, shift the origin from O to O_{new} . In the discrete case, consider a collection of charges q_i at positions $\boldsymbol{r}_i$ from O. The expression for $\boldsymbol{a}$ now becomes

$$\boldsymbol{a} = \frac{\sum_i q_i \boldsymbol{r}_i}{\sum_i q_i}.$$

Note: The denominator, that is, the total charge, should not be zero. When that happens, the dipole moment is independent of the choice of origin. So if it non-vanishing, it can never be made to vanish by a shift.

(c) Q is the quadrupole moment tensor.

$$Q_{ij} = \int \left(3r'_i r'_j - \left(r'^2\right) \delta_{ij}\right) \rho dV' \tag{2}$$

where δ_{ij} is Kronecker delta (Recall PH1010). $(Q_{xx})_{\text{new}}$ is the xx component of quadrupole moment tensor evaluated about O_{new} . Similarly, other components of Q_{new} are defined.

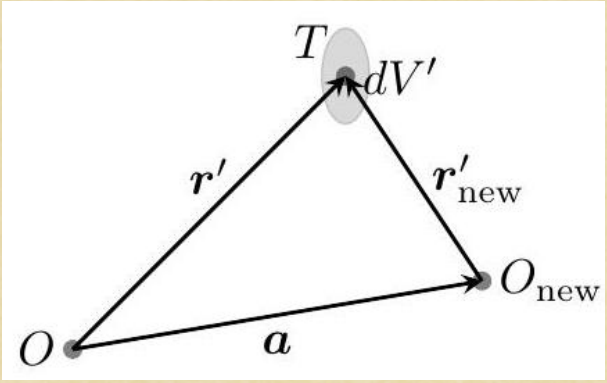


Figure 10: Finding a new origin about which the dipole moment vanishes.

$$\begin{aligned} x'_{\text{new}} &= x' - a_x \\ y'_{\text{new}} &= y' - a_y \\ z'_{\text{new}} &= z' - a_z \end{aligned} \tag{3}$$

$\boldsymbol{a} = a_x \hat{\mathbf{x}} + a_y \hat{\mathbf{y}} + a_z \hat{\mathbf{z}}$ is a constant vector. $(Q_{xx})_{\text{new}}$ (from Equation 2 above)

$$\int \rho(T) \left(2x'^2_{\text{new}} - y'^2_{\text{new}} - z'^2_{\text{new}}\right) dV' \tag{4}$$

Substitute 3 in 4 and regroup terms. Then,

$$
\begin{aligned}
(Q_{xx})_{\text{new}} &= \int \rho(T)\left(2x'^2 - y'^2 - z'^2\right) dV' \\
&\quad + \int \rho(T)\left(2a_x^2 - a_y^2 - a_z^2\right) dV' \\
&\quad + \int \left(-4x'a_x + 2y'a_y + 2z'a_z\right) dV'
\end{aligned} \tag{5}
$$

The first term in 5 is Q_{xx}, the xx component of the quadrupole moment tensor about O. The second term in 5 is

$$\left(2a_x^2 - a_y^2 - a_z^2\right) \int \rho dV'$$

where $q = \int \rho dV'$ is the total charge. The third term in 5 is

$$-4a_x \int \rho x' dV' + 2a_y \int \rho y' dV' + 2a_z \int \rho z' dV' = -4a_x p_x + 2a_y p_y + 2a_z p_z$$

This implies that if $\boldsymbol{p} = 0$ and $q = \int \rho dV' = 0$, then $(Q_{xx})_{\text{new}} = Q_{xx}$. More generally, one can show that

$$(Q_{ij})_{\text{new}} = Q_{ij} - 3a_i p_j - 3a_j p_i + (\mathbf{a}\cdot\mathbf{p})\delta_{ij} + 3qa_i a_j - qa^2\delta_{ij}$$

where $\mathbf{p}$ and q are the dipole moment and the total charge respectively in the original coordinate system. Thus, if $\mathbf{p} = 0$ and $q = 0$, we see that $Q_{ij})_{\text{new}} = Q_{ij}$.

7. Problem-7

Question: Show that for a spherically symmetric charge distribution, the dipole, quadrupole, and all higher moments about the center of the distribution are identically zero.
Solution:

Consider a spherical distribution of charges with total charge Q. At a point $\boldsymbol{r}$ outside the distribution,

$$\phi(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0 r} \int \rho dV.$$

But this is the monopole term in the multipole expansion. This implies that all higher moments vanish.

8. Problem-8

Question: Three point charges $+q$, $+2q$, and $-4q$ are held fixed at points with Cartesian coordinates $(0, 0, d)$, $(-d, 0, d)$, and (d, d, d) respectively, in a Cartesian coordinate system. Calculate the quadrupole moment of the charge distribution about the point where the dipole moment vanishes.

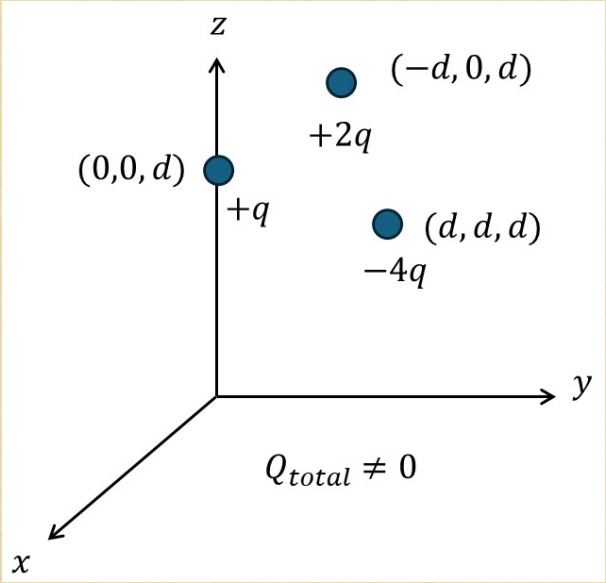


Figure 11: Three point charges $+q$, $+2q$, and $-4q$ are held fixed at points with Cartesian coordinates $\boldsymbol{r_1^0}$ $(0, 0, d)$, $\boldsymbol{r_2^0}$ $(-d, 0, d)$, and $\boldsymbol{r_3^0}$ (d, d, d), respectively

Solution:

From Problem (6b), the point about which the dipole moment of the charge distribution is zero, has position vector

$$\boldsymbol{a} = \frac{\sum_i q_i \boldsymbol{r}_i}{\sum_i q_i}$$

This point is called the **centre of charge**. Therefore, in this problem the x component of the position vector of the center of charge is $6d$. Similarly, the y coordinate is $4d$, and the z coordinate is d. Thus, the position vector for the centre of charge is $(6d, 4d, d)$. With respect to this centre of charge,

(a) charge q has position vector $\boldsymbol{r}_1 = \boldsymbol{r_1^0} - \boldsymbol{a} = (-6d, -4d, 0)$

(b) charge $2q$ has position vector $\boldsymbol{r}_2 = \boldsymbol{r_2^0} - \boldsymbol{a} = (-7d, -4d, 0)$

(c) charge $-4q$ has position vector $\boldsymbol{r}_3 = \boldsymbol{r_3^0} - \boldsymbol{a} = (-5d, -3d, 0)$

$$\therefore Q_{xx} = \sum_{i=1}^{3} \left(3x_i^2 - r_i^2\right)$$

Note:

(a) $x_1 = -6d$ and $\boldsymbol{r}_i^2 = 52d^2$ and so on. Substituting and simplifying, we get $Q_{xx} = 56qd^2$. Similarly, Q_{yy} and Q_{zz} can be obtained. (b) $Q_{zz} = -(Q_{xx} + Q_{yy})$. While $Q_{xy} = Q_{yx} = 60qd^2$, all other components of Q are zero because the z coordinate of all the charges is zero.

9. Problem-9

Question: Find the monopole, dipole, and quadrupole moments about the origin for the following charge distributions:

(a) A line charge of constant charge density λ lying in the first quadrant of the xy-plane with one end at the origin and making an angle φ_0 with the positive x-axis (see Fig. 12).

(b) A solid cylinder of height H and radius R with its geometrical center at the origin, having a uniform volume charge density ρ_0, and its axis along the z-axis.

Solution:

(a) Monopole moment $q = \lambda L$.

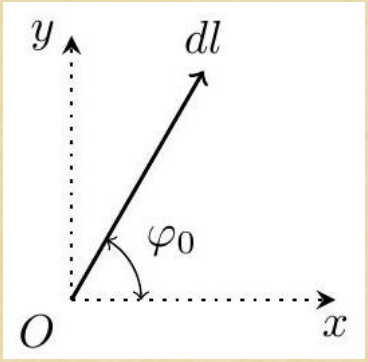


Figure 12: A line charge (of density λ) lying in the first quadrant of the xy plane, making an angle φ_0 with positive x axis

Dipole moment about O is $\boldsymbol{p} = \int_0^L \boldsymbol{r}\lambda dl$

$$\boldsymbol{r} = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} \text{ with } x = l\cos\varphi_0, y = l\sin\varphi_0$$

Substituting for x and y in terms of l

$$\begin{aligned}\boldsymbol{p} &= \lambda \int_0^L (l\cos\varphi_0\hat{\mathbf{x}} + l\sin\varphi_0\hat{\mathbf{y}})\, dl \\ &= \frac{\lambda L^2}{2}(\cos\varphi_0\hat{\mathbf{x}} + \sin\varphi_0\hat{\mathbf{y}})\end{aligned}$$

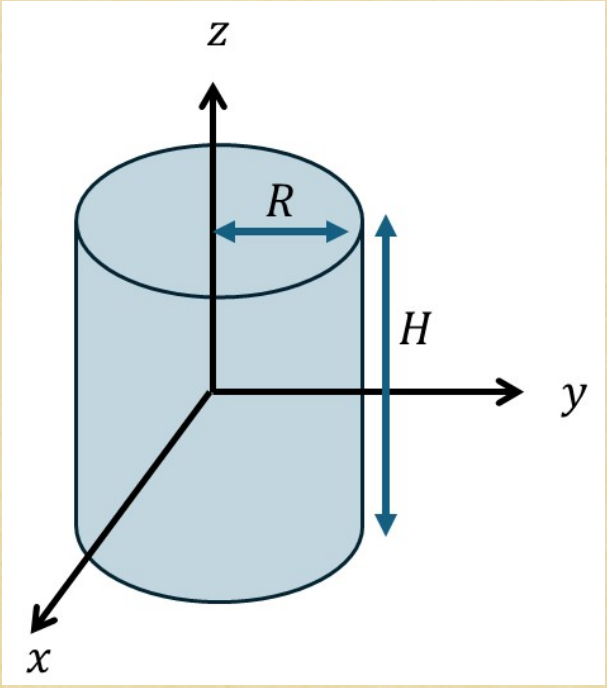


Figure 13: solid cylinder of height H and radius R with its geometrical center at the origin

$$
\begin{aligned}
Q_{xx} &= \lambda \int_0^L \left(3x^2 - r^2\right) dl \\
&= \lambda \int_0^L \left(3l^2 \cos^2\varphi_0 - l^2\right) dl \\
&= \left(3\cos^2\varphi_0 - 1\right) \frac{\lambda L^3}{3} \\
Q_{yy} &= \lambda \int_0^L \left(3y^2 - r^2\right) dl \\
&= \left(3\sin^2\varphi_0 - 1\right) \frac{\lambda L^3}{3} \\
Q_{zz} &= -\left(Q_{xx} + Q_{yy}\right) \\
&= -\frac{\lambda L^3}{3} \\
Q_{xy} &= 3\lambda \cos\varphi_0 \sin\varphi_0 \int_0^L l^2 dl \\
&= \lambda L^3 \cos\varphi_0 \sin\varphi_0 \\
Q_{yz} &= Q_{xz} = 0. \quad (\because z \text{ coordinate is zero })
\end{aligned}
$$

In summary we have $q = \lambda L$,

$$
\mathbf{p} = \frac{\lambda L^2}{2}\left(\cos\varphi_0 \hat{\mathbf{x}} + \sin\varphi_0 \hat{\mathbf{y}}\right) \quad , \quad Q = \lambda L^3 \begin{pmatrix} \cos^2\varphi_0 - \frac{1}{3} & \cos\varphi_0 \sin\varphi_0 & 0 \\ \cos\varphi_0 \sin\varphi_0 & \sin^2\varphi_0 - \frac{1}{3} & 0 \\ 0 & 0 & -\frac{1}{3} \end{pmatrix}.
$$

(b) Limits of integration: $-H/2 \le z' \le H/2, 0 \le \varrho' \le R$ and $0 \le \varphi' < 2\pi$.

$$\text{Monopole term} \;=\; \text{total charge } q = \rho_0 \pi R^2 H$$

Dipole moment about origin $= 0 \quad (\because$ reflection symmetry $)$

$$
\begin{aligned}
Q_{xx} &= Q_{yy} \quad (\because x \text{ and } y \text{ are equivalent }) \\
Q_{zz} &= \int \left(3z'^2 - r'^2\right) \rho_0 dV' \quad r'^2 = x'^2 + y'^2 + z'^2 \\
dV' &= \varrho' d\varrho' d\varphi dz \quad \varrho'^2 = x'^2 + y'^2 \\
\therefore Q_{zz} &= \rho_0 \int \left(2z'^2 - \rho'^2\right) \varrho' d\varrho' d\varphi' dz' \\
&= \frac{\rho_0 \pi H R^2}{6} \left(H^2 - 3R^2\right) \\
&= \frac{q}{6} \left(H^2 - 3R^2\right) \\
&= -\left(Q_{xx} + Q_{yy}\right) \\
Q_{yz} &= 0 \quad (\because \rho(x, y, z) = \rho(x, y, -z)) \\
\text{Similarly, } Q_{xy} &= Q_{xz} = 0.
\end{aligned}
$$

In summary we have

$$
q = \rho_0 \pi R^2 H \quad , \quad \mathbf{p} = 0 \quad , \quad Q = \frac{q\left(H^2 - 3R^2\right)}{12} \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix}.
$$

[1]

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